Object Detection

AZURE CUSTOM VISION

Objective

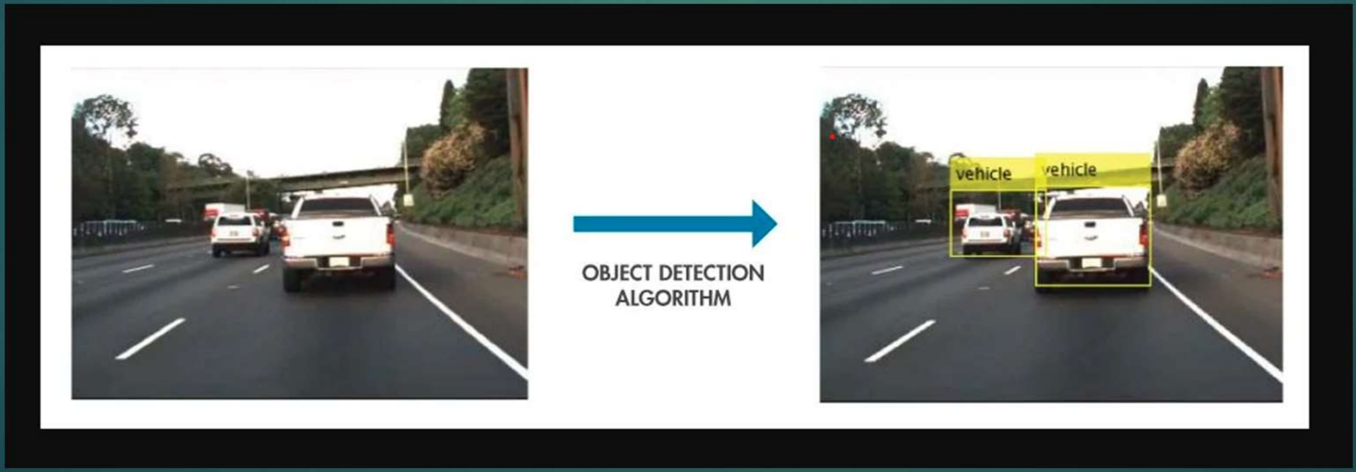
- Understanding how to develop an API endpoint using Microsoft Azure Custom Vision for object detection.

Domain

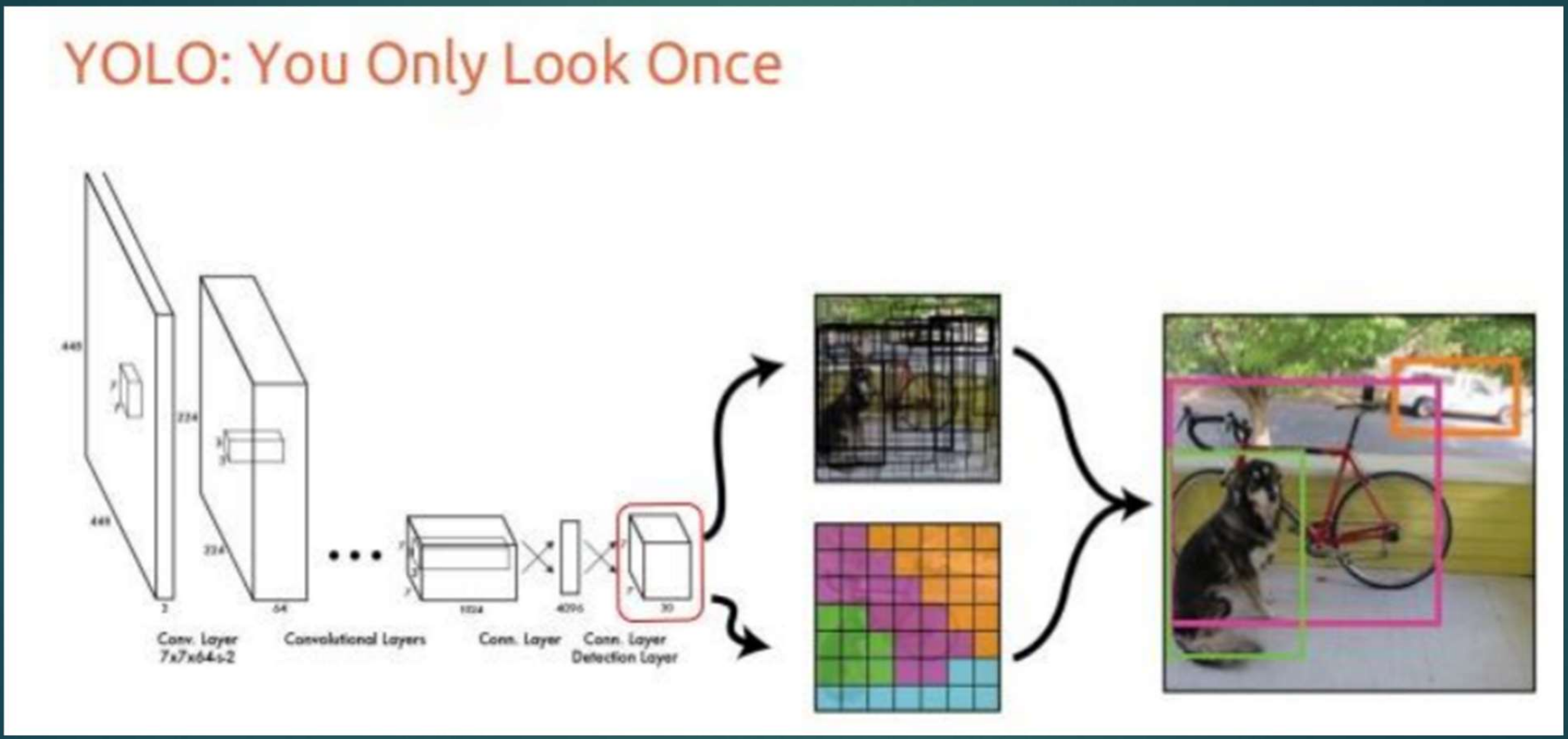
- Humans can easily identify and detect object present in an image. By leveraging vision capabilities present using vision models, we can automate this process.
- The real-world applications of object detection can be seen in many crucial areas of our lives, such as medical imaging, video tracking, movement detection, facial recognition, and object recognition, even in autonomous vehicles.

Object Detection

- Object detection is a computer vision technique that aims to locate and identify objects within an image or video. Object detection also allows the detection of objects or targets from a visual feed in real time. The process involves steps designed to recognize and label the distinct entities in the graphical data.



Algorithm Architecture (Basic)



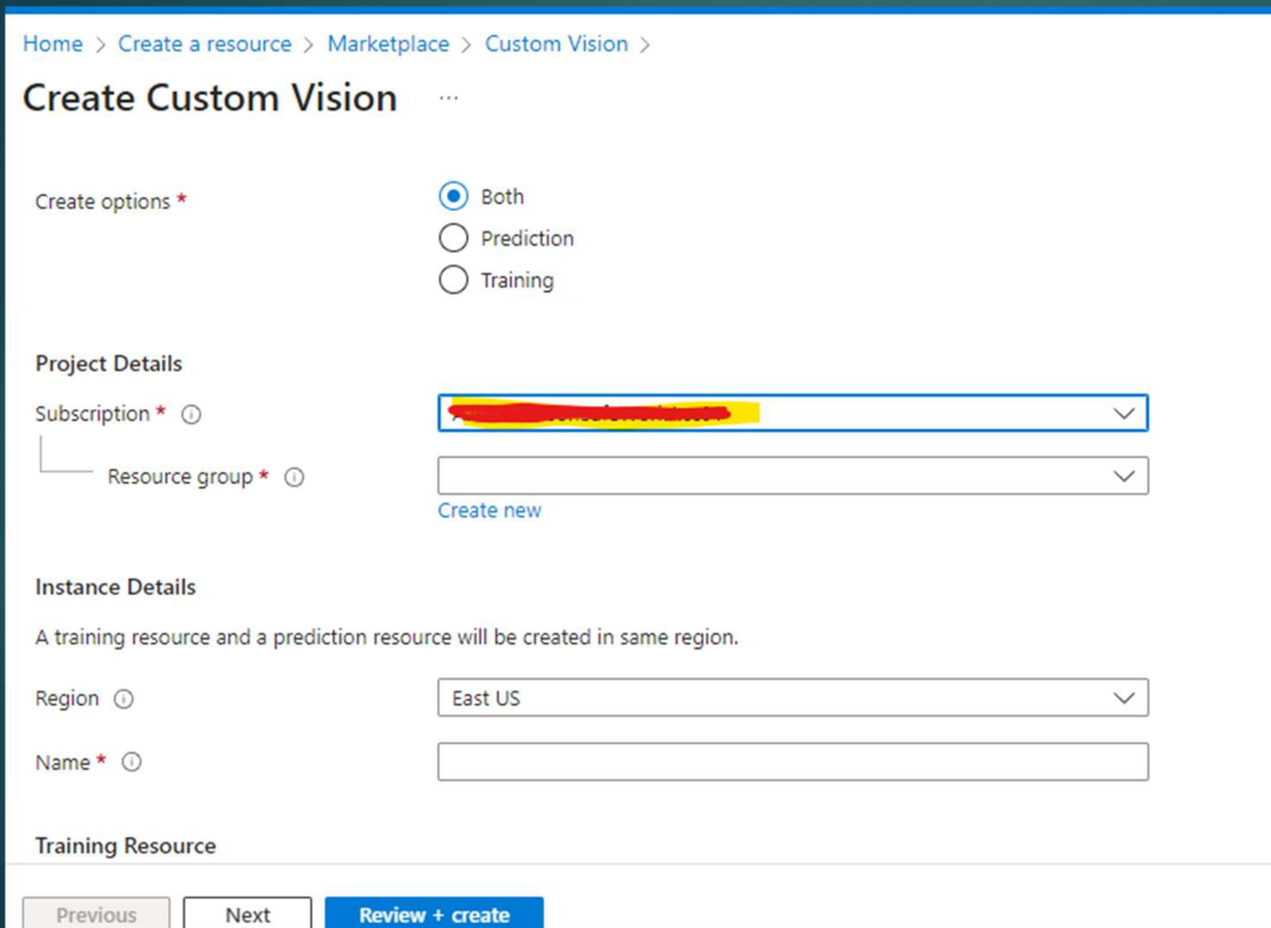
Custom Vision

- Azure AI Custom Vision is an image recognition service that lets you build, deploy, and improve your own object detection and classification models.
- The Custom Vision service uses a machine learning algorithm to analyze images for custom features.
- Custom Vision functionality can be divided into two features. Classification applies one or more labels to an entire image. Detection is similar, but it returns the coordinates in the image where the applied labels are found.

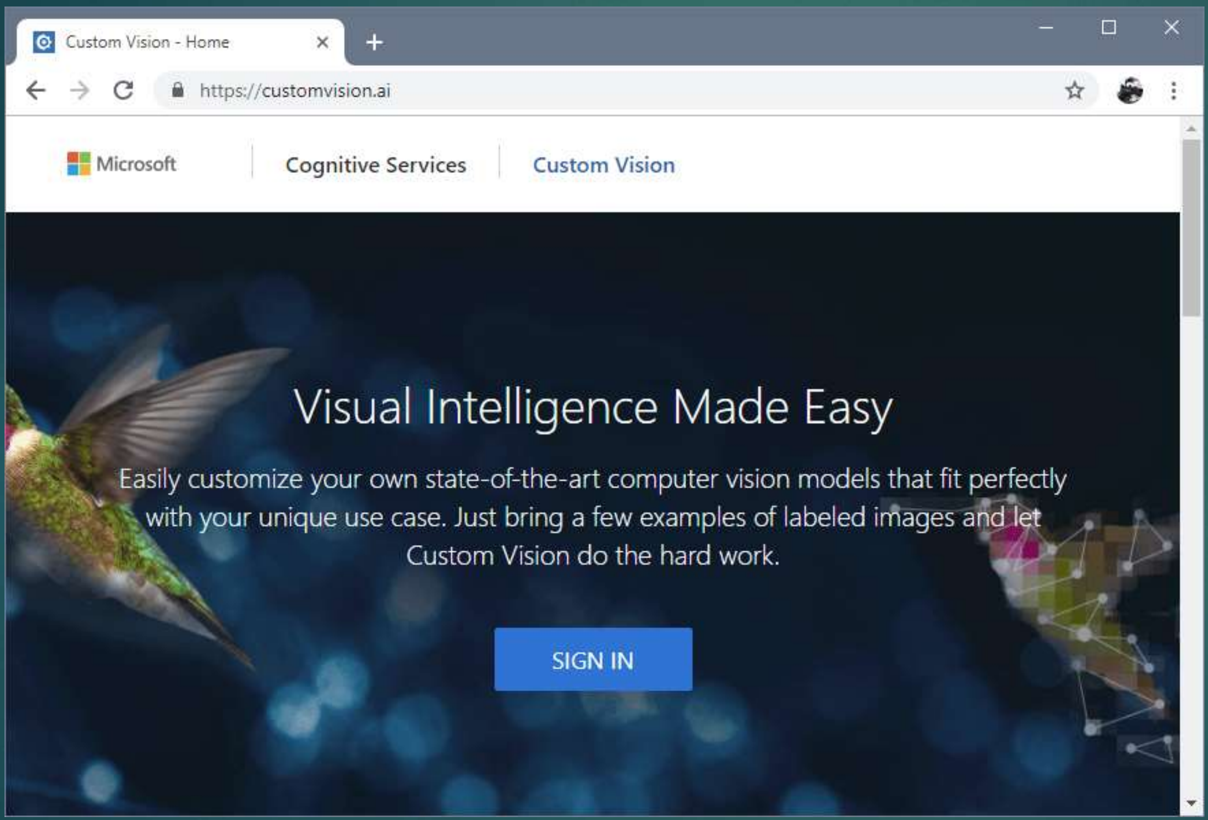
Prerequisites

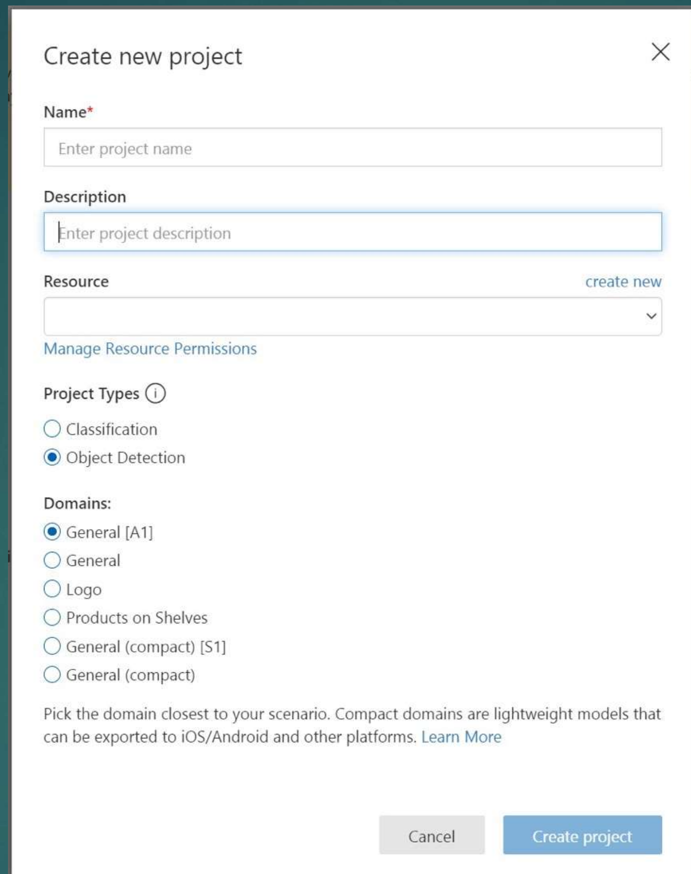
- Azure subscription or create a free account.
- Need to create a custom vision resource in https://portal.azure .com
- A set of images or sample images downloaded from interent.
- Supported Web Browser (Chrome, Bing)

Create Resource

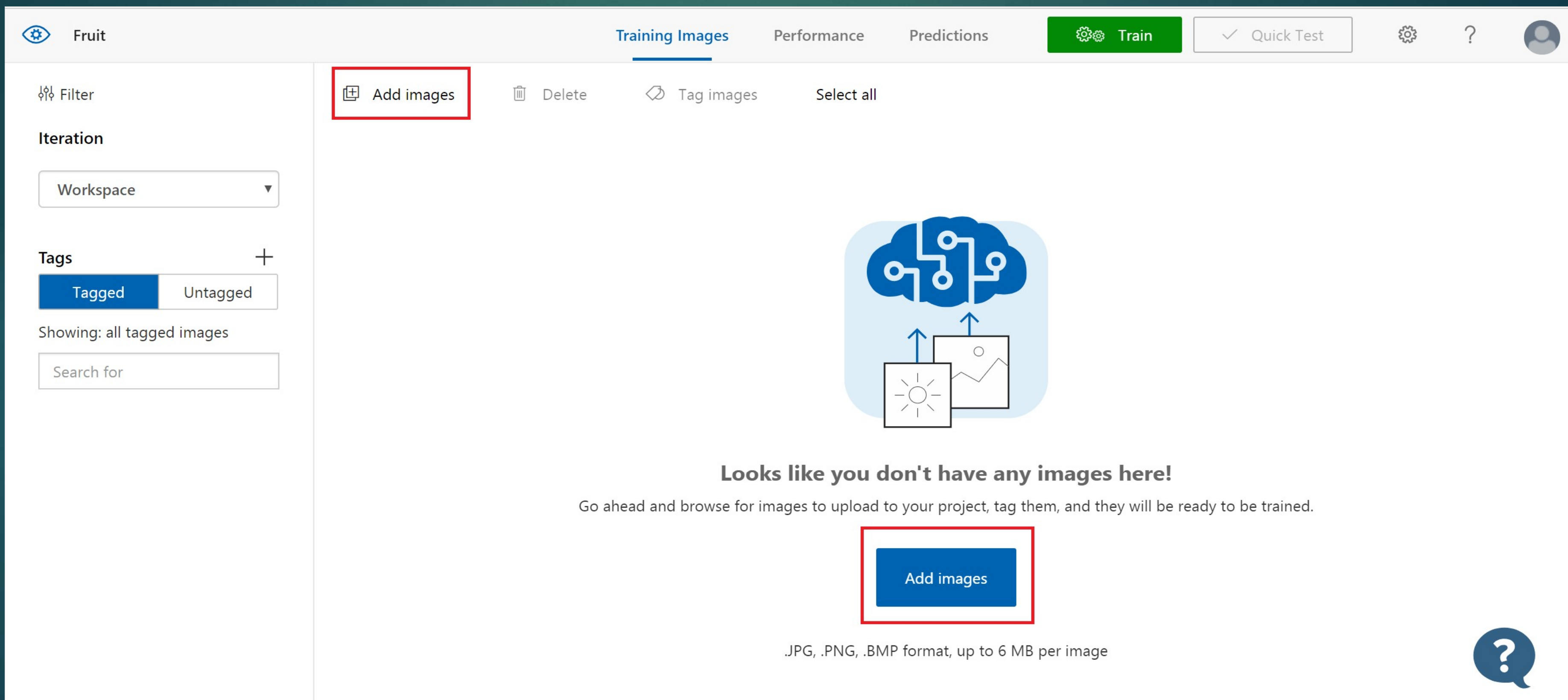


Create a project




Create new project
Name*
Enter project name
Description
Enter project description
Resource
create new
Manage Resource Permissions
Project Types
Classification
Object Detection
Domains:
General [A1]
General
Logo
Products on Shelves
General (compact) [S1]
General (compact)
Pick the domain closest to your scenario. Compact domains are lightweight models that can be exported to iOS/Android and other platforms. Learn More
Cancel
Create project

Upload and tag images



Tagging

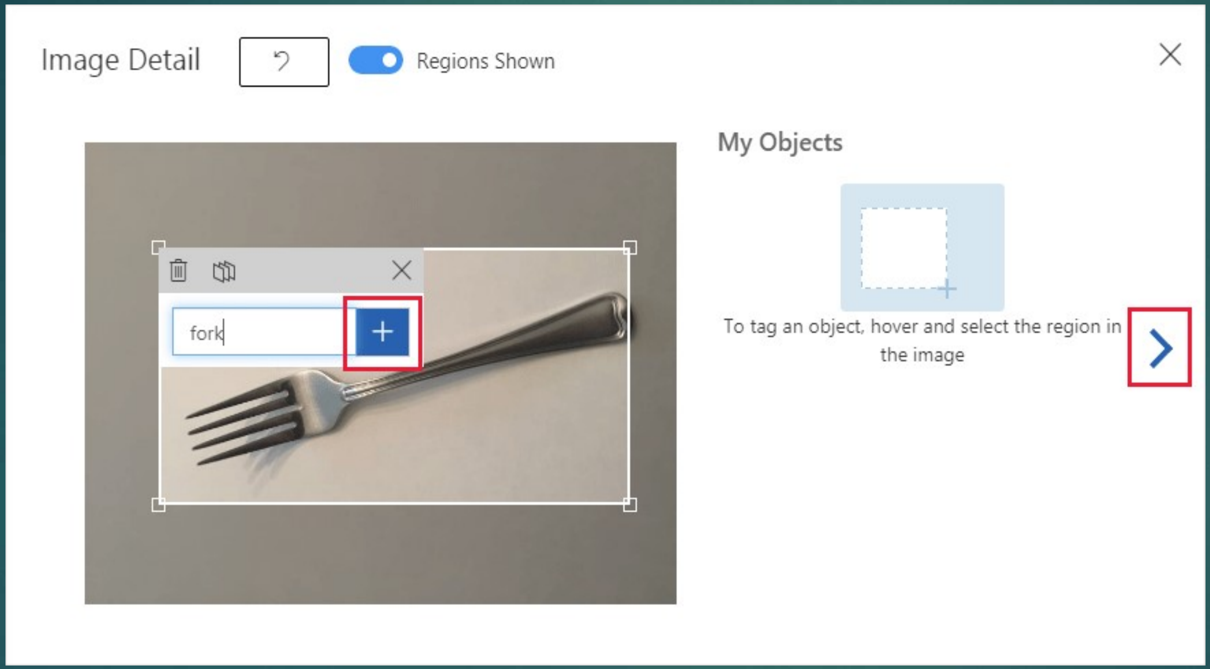
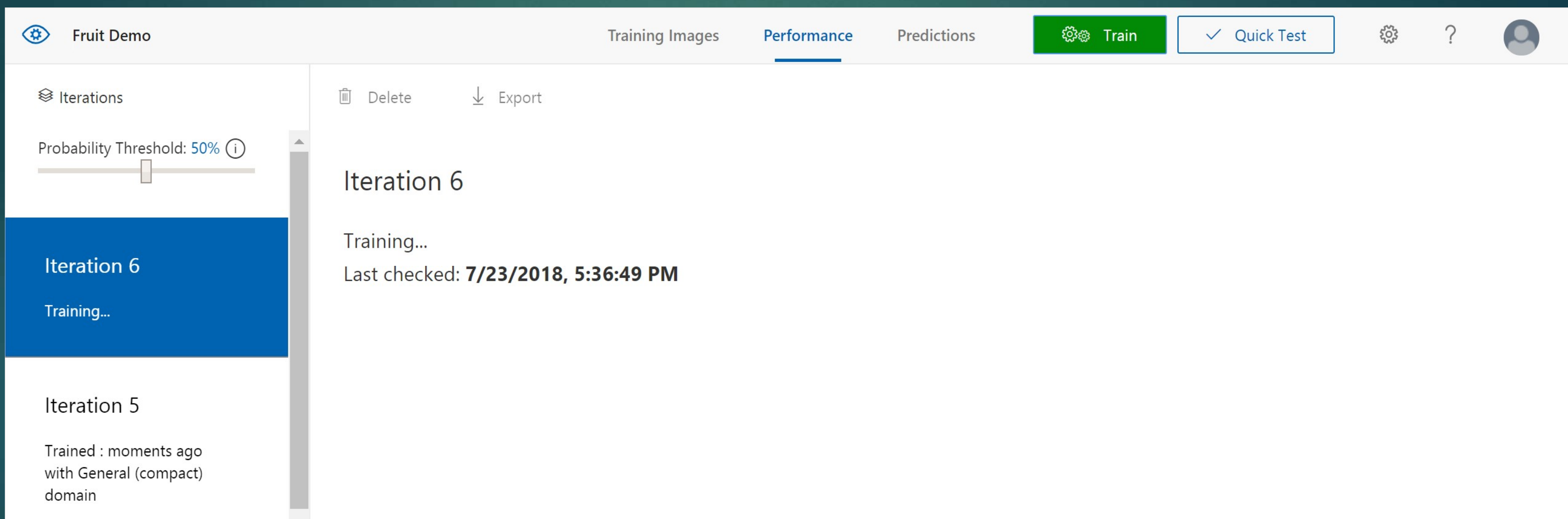
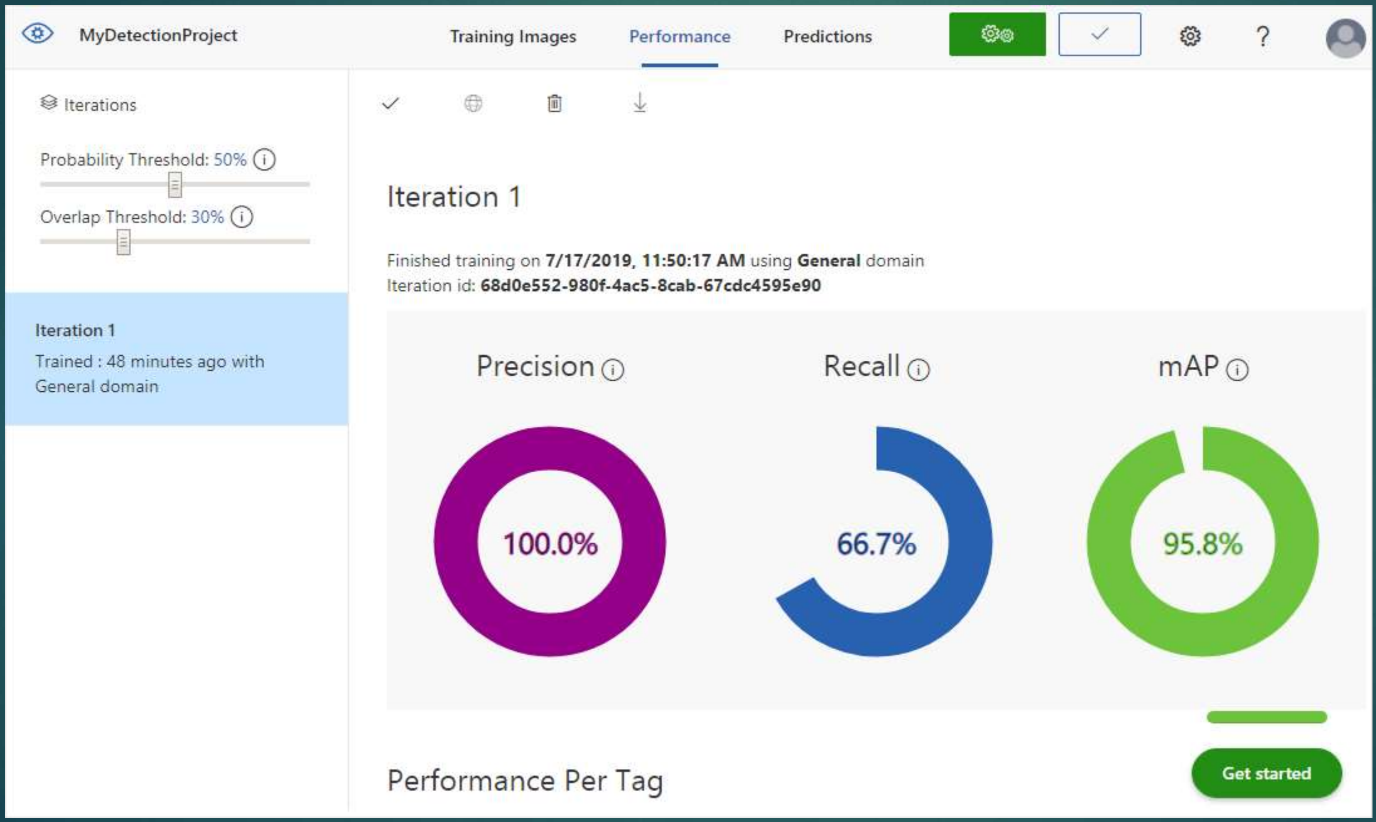


Image Detail
Regions Shown
fork
My Objects
To tag an object, hover and select the region in the image

Train images



Evaluation



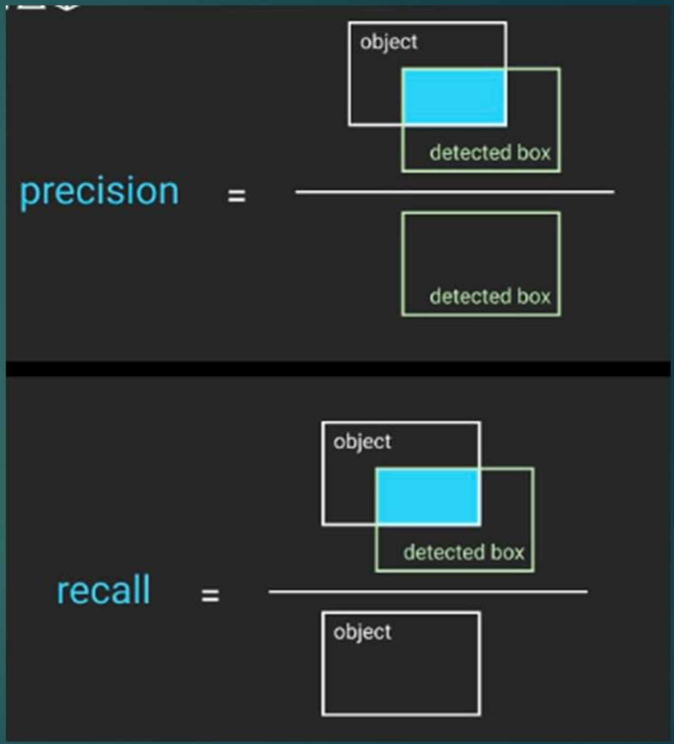
MyDetectionProject
Training Images
Performance
Predictions
Iterations
Probability Threshold: 50%
Overlap Threshold: 30%
Iteration 1
Trained : 48 minutes ago with General domain
Iteration 1
Finished training on 7/17/2019, 11:50:17 AM using General domain
Iteration id: 68d0e552-980f-4ac5-8cab-67cdc4595e90
Precision
Recall
mAP
100.0%
66.7%
95.8%
Performance Per Tag
Get started

Metrics

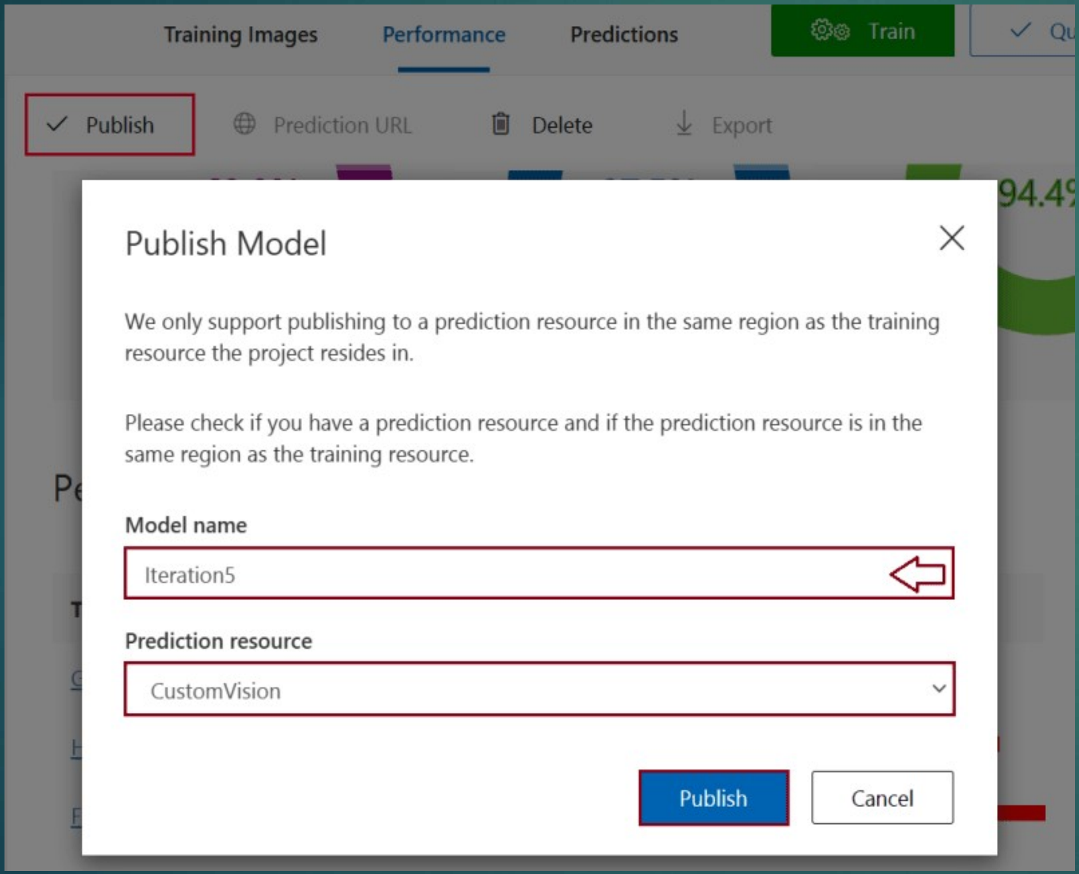
- Precision = Precision measures the accuracy of positive predictions.

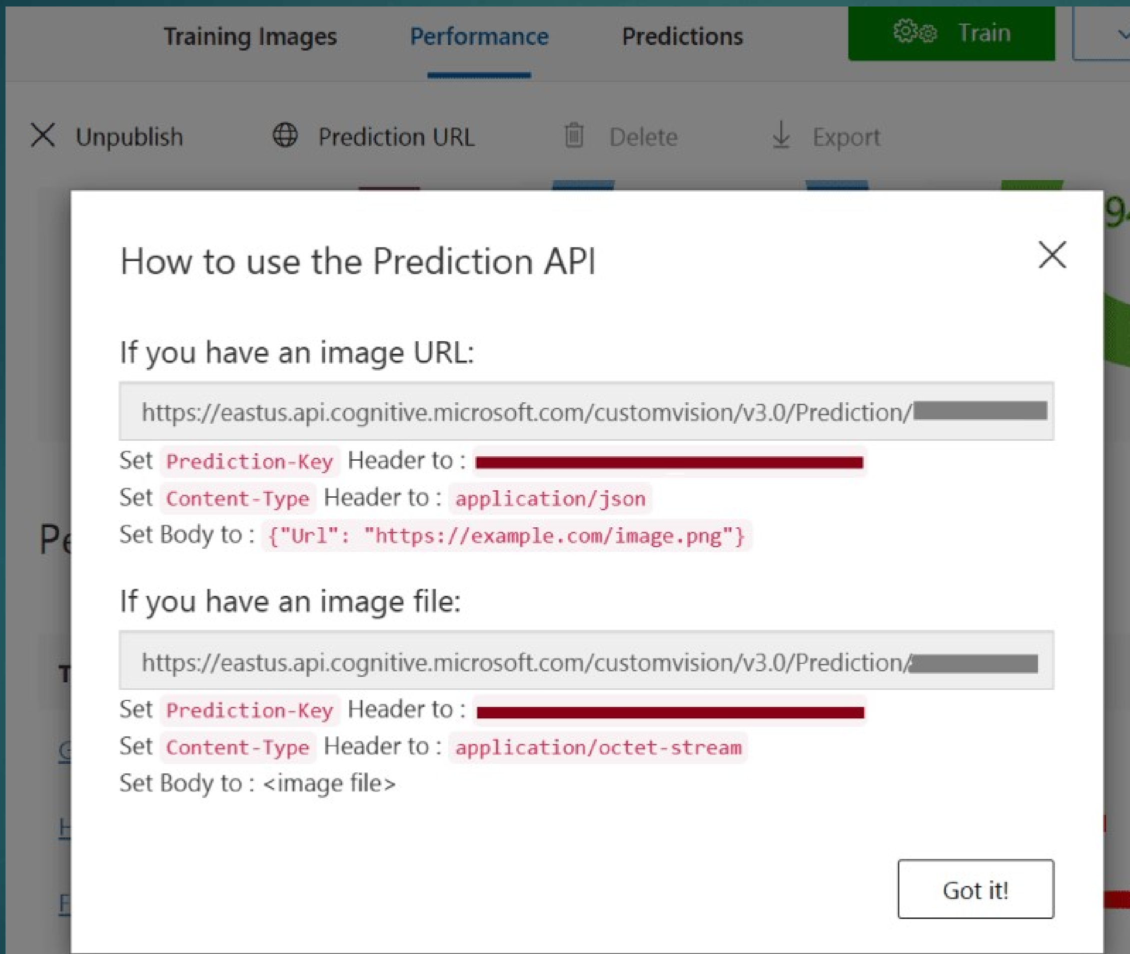
  (True Positive)/(True Positive + False Positive)

- Recall = Measures the ability to identify all relevant events.(True Positive)/(True Positive + False Negative)
- Mean Average Precision (mAP) provides a comprehensive evaluation,especially in multi –class object detection tasks.



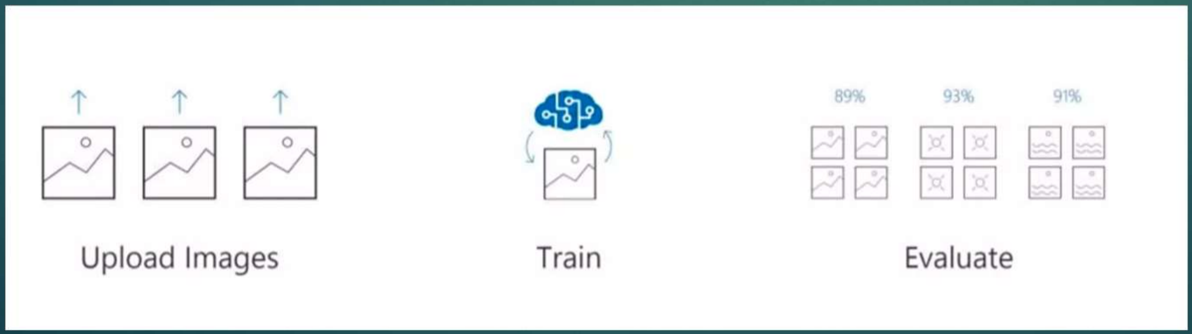
Publishing


Training Images
Performance
Predictions
Train
Publish
Prediction URL
Delete
Export
Publish Model
We only support publishing to a prediction resource in the same region as the training resource the project resides in.
Please check if you have a prediction resource and if the prediction resource is in the same region as the training resource.
Model name
Iteration5
Prediction resource
CustomVision
Publish
Cancel


Training Images
Performance
Predictions
Train
Unpublish
Prediction URL
Delete
Export
How to use the Prediction API
If you have an image URL:
https://eastus.api.cognitive.microsoft.com/customvision/v3.0/Prediction/[redacted]
Set Prediction-Key Header to : [redacted]
Set Content-Type Header to : application/json
Set Body to : {"Url": "https://example.com/image.png"}
If you have an image file:
https://eastus.api.cognitive.microsoft.com/customvision/v3.0/Prediction/[redacted]
Set Prediction-Key Header to : [redacted]
Set Content-Type Header to : application/octet-stream
Set Body to : <image file>
Got it!

Offline Model

- Export the trained model for Android,iOs, Docker image etc.
- The same can be achieved to Azure Python SDK
- Steps:
  - Install the necessary packages.
  - Load the appropriate key values.
  - Perform the prediction



Important tips

- Minimum images per tag is 15.
- Object should be the focus of the picture.
- Diverse images for better training and recall value.
- Image size should not be more than 6MB.
- The image aspect ratio should not be larger than 25:1.

References

- https://learn.microsoft.com/en-us/azure/ai-services/custom-vision-service/getting-started-build-a-classifier
- https://learn.microsoft.com/en-us/azure/ai-services/custom-vision-service/concepts/compare-alternatives
- https://labelyourdata.com/articles/object-detection-metrics#:~:text=The%20performance%20of%20an%20object,mAP)%20across%20different%20object%20categories.

Thank you.